

Mechanics



Lecture 10
Rigid Body Motion
(Chapter 5)

What We Did Last Time

- Found the velocity due to rotation

- Used it to find Coriolis effect

$$\left(\frac{d}{dt}\right)_s = \left(\frac{d}{dt}\right)_r + \boldsymbol{\omega} \times$$

- Connected $\boldsymbol{\omega}$ with the Euler angles

- Lagrangian $\rightarrow$ translational and rotational parts

- Often possible if body axes are defined from the CoM

- Defined the inertia tensor

$$I_{jk} = m_i (r_i^2 \delta_{jk} - x_{ij} x_{ik})$$

- Calculated angular momentum and kinetic energy

$$\mathbf{L} = \mathbf{I}\boldsymbol{\omega}$$

$$T = \frac{\boldsymbol{\omega} \cdot \mathbf{I}\boldsymbol{\omega}}{2} = \frac{1}{2} I \omega^2$$

Diagonalizing Inertia Tensor

$$\mathbf{I} = \begin{bmatrix} m_i(r_i^2 - x_i^2) & -m_i x_i y_i & -m_i x_i z_i \\ -m_i y_i x_i & m_i(r_i^2 - y_i^2) & -m_i y_i z_i \\ -m_i z_i x_i & -m_i z_i y_i & m_i(r_i^2 - z_i^2) \end{bmatrix} = m_i (r_i^2 \delta_{jk} - x_{ij} x_{ik})$$

- Inertia tensor I can be made diagonal

$$\mathbf{I}_D = \mathbf{R} \mathbf{I} \tilde{\mathbf{R}} = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}$$

$\mathbf{R}$ is a rotation matrix

$$I_i > 0$$

Kinematical properties of a rigid body are fully described by its mass, principal axes, and moments of inertia

Principal Axes

- Consider a rigid body with body axes x - y - z
 - Inertia tensor $\mathbf{I}$ is (in general) not diagonal
 - But it can be made diagonal by $\mathbf{I}_D = \mathbf{R}\mathbf{I}\tilde{\mathbf{R}}$
- Rotate x - y - z by $\mathbf{R} \rightarrow$ New body axes x' - y' - z'
 - In x' - y' - z' coordinates

$$\boldsymbol{\omega}' = \mathbf{R}\boldsymbol{\omega}$$

$$\begin{aligned}\mathbf{L}' &= \mathbf{R}\mathbf{L} \\ &= \mathbf{R}\mathbf{I}\boldsymbol{\omega} \\ &= \mathbf{R}\mathbf{I}\tilde{\mathbf{R}}\mathbf{R}\boldsymbol{\omega} \\ &= \mathbf{I}_D\boldsymbol{\omega}'\end{aligned}$$

$$\tilde{\mathbf{R}}\mathbf{R} = \mathbf{1}$$

One can choose a set of body axes that make the inertia tensor diagonal
 $\rightarrow$ **Principal Axes**

How do I find them?

Finding Principal Axes

$$\mathbf{I}_D = \mathbf{R}\mathbf{I}\tilde{\mathbf{R}} = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}$$

- Consider unit vectors $\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3$ along principal axes

$$\mathbf{I}\mathbf{n}_i = I_i\mathbf{n}_i$$

← $\mathbf{n}_i$ is an eigenvector of $\mathbf{I}$ with eigenvalue I_i

- To find the principal axes and principal moments:
 - Express $\mathbf{I}$ in any body coordinates
 - Solve eigenvalue equation
$$(\mathbf{I} - \lambda)\mathbf{r} = 0 \Rightarrow |\mathbf{I} - \lambda| = 0 \Rightarrow \lambda = I_1, I_2, I_3$$
 - Eigenvectors point the principal axes
 - Use them to re-define the body coordinates to simplify $\mathbf{I}$
- You can often find the principal axes by just looking at the object

Rotational Equation of Motion

- Concentrate on the rotational motion

$$\left(\frac{d}{dt}\right)_s = \left(\frac{d}{dt}\right)_b + \boldsymbol{\omega} \times$$

- Newtonian equation of motion gives

$$\left(\frac{d\mathbf{L}}{dt}\right)_s = \mathbf{N} \quad \longrightarrow \quad \left(\frac{d\mathbf{L}}{dt}\right)_b + \boldsymbol{\omega} \times \mathbf{L} = \mathbf{N}$$

“space” axes“body” axes

- Take the **principal axes** as the body axes

$$\mathbf{L} = \mathbf{I}\boldsymbol{\omega} = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \begin{bmatrix} I_1\omega_1 \\ I_2\omega_2 \\ I_3\omega_3 \end{bmatrix}$$

Euler's Equation of Motion

$$\frac{d}{dt} \begin{bmatrix} I_1 \omega_1 \\ I_2 \omega_2 \\ I_3 \omega_3 \end{bmatrix} + \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \times \begin{bmatrix} I_1 \omega_1 \\ I_2 \omega_2 \\ I_3 \omega_3 \end{bmatrix} = \begin{bmatrix} N_1 \\ N_2 \\ N_3 \end{bmatrix} \quad \Rightarrow \quad \begin{aligned} I_1 \dot{\omega}_1 - \omega_2 \omega_3 (I_2 - I_3) &= N_1 \\ I_2 \dot{\omega}_2 - \omega_3 \omega_1 (I_3 - I_1) &= N_2 \\ I_3 \dot{\omega}_3 - \omega_1 \omega_2 (I_1 - I_2) &= N_3 \end{aligned}$$

Euler's equation of motion for rigid body
with one point fixed

- Special cases:

- $\omega_2 = \omega_3 = 0 \quad \Rightarrow \quad I_1 \dot{\omega}_1 = N_1$

- $I_2 = I_3 \quad \Rightarrow \quad I_1 \dot{\omega}_1 = N_1$

Torque-Free Motion

- No linear force $\rightarrow$ Conservation of linear momentum
- No torque $\rightarrow$ Conservation of angular momentum
- Try $\mathbf{N} = 0$ in Euler's equation of motion

$$I_1 \dot{\omega}_1 - \omega_2 \omega_3 (I_2 - I_3) = 0$$

$$I_2 \dot{\omega}_2 - \omega_3 \omega_1 (I_3 - I_1) = 0$$

$$I_3 \dot{\omega}_3 - \omega_1 \omega_2 (I_1 - I_2) = 0$$

Integrating these equations will give us energy and angular momentum conservation

- We will do something more intuitive (hopefully)
 - Geometrical trick by L. Poinsot

Inertia Ellipsoid

- For any direction $\mathbf{n}$, $I = \mathbf{n} \cdot \mathbf{I} \mathbf{n} = n_i I_{ij} n_j$
 - If we express $\mathbf{n}$ using principal axes $x'-y'-z'$

Moment of inertia
about axis $\mathbf{n}$

$$I = I_i n_i^2 = I_1 n_1^2 + I_2 n_2^2 + I_3 n_3^2$$

- Consider a vector $\boldsymbol{\rho} = \frac{\mathbf{n}}{\sqrt{I}}$

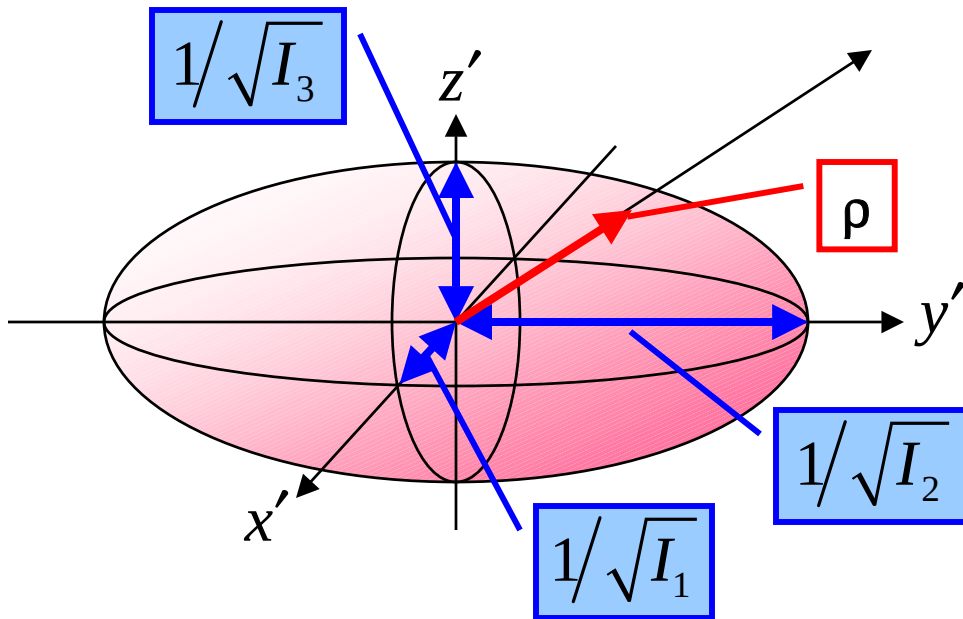
$$\Rightarrow 1 = I_i \rho_i^2 = I_1 \rho_1^2 + I_2 \rho_2^2 + I_3 \rho_3^2$$

Inertia ellipsoid

Inertia Ellipsoid

$$\boldsymbol{\rho} = \frac{\mathbf{n}}{\sqrt{I}}$$

$$1 = I_1 \rho_1^2 + I_2 \rho_2^2 + I_3 \rho_3^2$$



Inertia ellipsoid represents the moment of inertia of a rigid body in all directions

Usefulness of this definition will become apparent soon ...

Inertia Ellipsoid

$$\boldsymbol{\rho} = \frac{\mathbf{n}}{\sqrt{I}}$$

- Inertia along axis $\mathbf{n}$ is $I = \mathbf{n} \cdot \mathbf{I} \mathbf{n}$

$$\Rightarrow F(\boldsymbol{\rho}) \equiv \boldsymbol{\rho} \cdot \mathbf{I} \boldsymbol{\rho} = I_1 \rho_1^2 + I_2 \rho_2^2 + I_3 \rho_3^2 = 1$$

- F is a function (like potential) defined in the $\boldsymbol{\rho}$ space

- $F(\boldsymbol{\rho}) = 1 \rightarrow$ Inertia ellipsoid

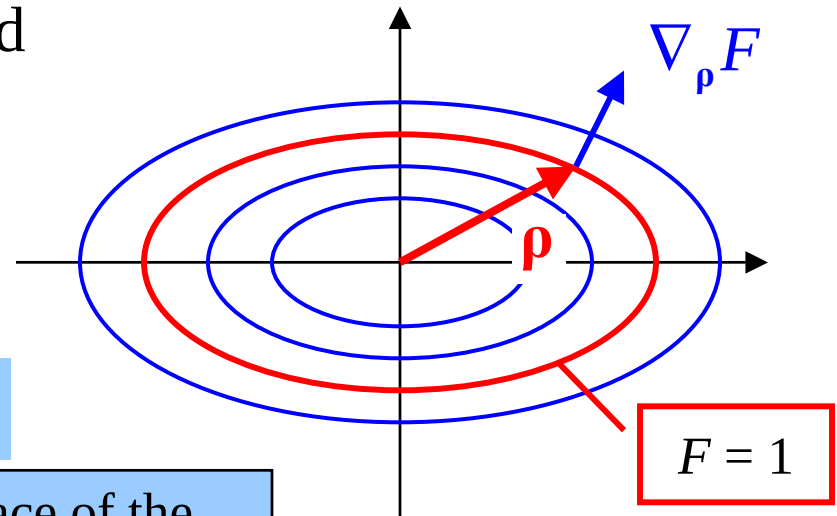
- Normal of the ellipsoid given by the gradient

$$\nabla_{\boldsymbol{\rho}} F = 2\mathbf{I} \boldsymbol{\rho}$$

- Using $\boldsymbol{\rho} = \mathbf{n} / \sqrt{I} = \boldsymbol{\omega} / \sqrt{2T}$

$$\nabla_{\boldsymbol{\rho}} F = \sqrt{\frac{2}{T}} \mathbf{L}$$

Surface of the inertia ellipsoid is perpendicular to $\mathbf{L}$



Invariable Plane

L is conserved!

$$\nabla_{\rho} F = \sqrt{\frac{2}{T}} \mathbf{L}$$

Surface of inertia ellipsoid at ρ is perpendicular to the angular momentum $\mathbf{L}$

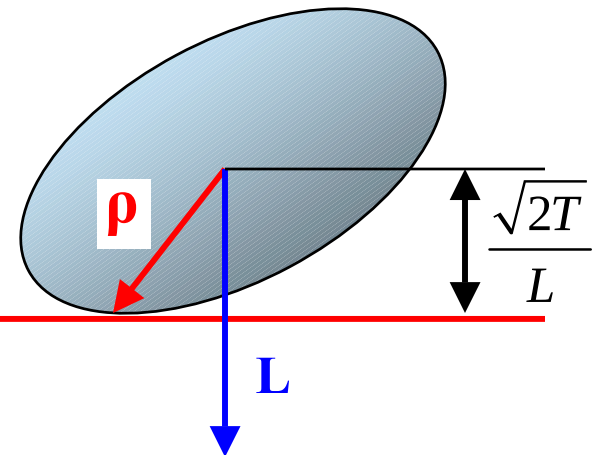
- As ρ moves, inertia ellipsoid must rotate to satisfy this condition continuously
- Consider the projection of ρ on $\mathbf{L}$

$$\frac{\rho \cdot \mathbf{L}}{L} = \frac{\omega \cdot \mathbf{L}}{L\sqrt{2T}} = \frac{\sqrt{2T}}{L}$$

constant

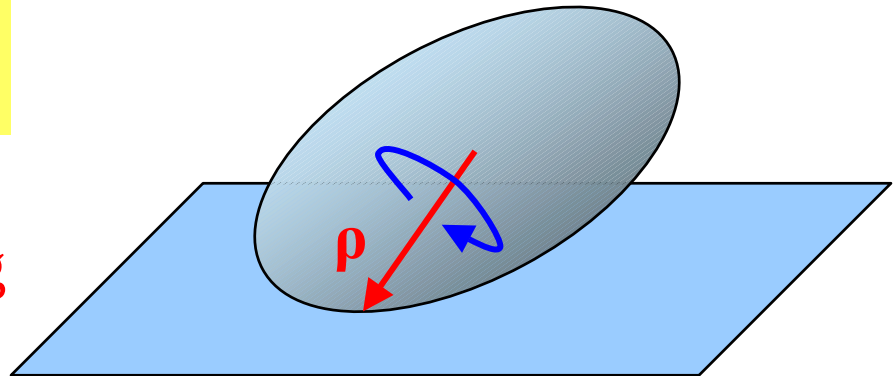
- The ellipsoid is touching a fixed plane perpendicular to $\mathbf{L}$

Invariable plane



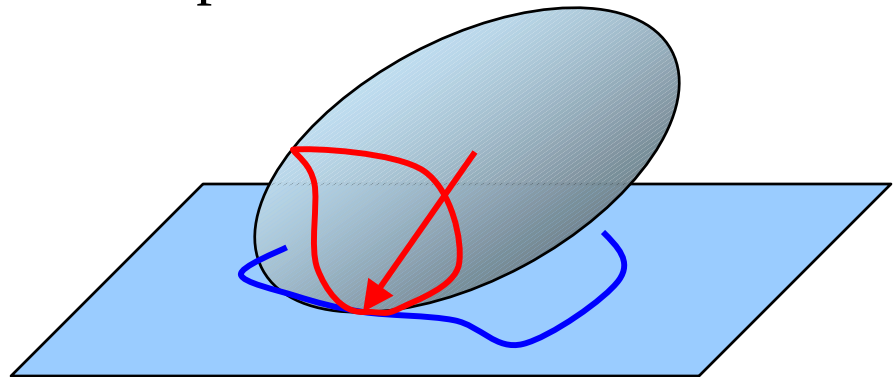
Invariable Plane

- Inertia ellipsoid touches the invariable plane
 - Distance between the center and the plane is $\sqrt{2T}/L$
 - Determined by the initial conditions
- Touching point = instantaneous axis of rotation
 - Tip of the $\boldsymbol{\rho}$ vector is momentarily at rest in space
 - $$\frac{d\boldsymbol{\rho}}{dt} = \boldsymbol{\omega} \times \boldsymbol{\rho} = \frac{\boldsymbol{\omega} \times \boldsymbol{\omega}}{\sqrt{2T}} = 0$$
 - i.e. it's not sliding, but **rolling without slipping** on the invariable plane



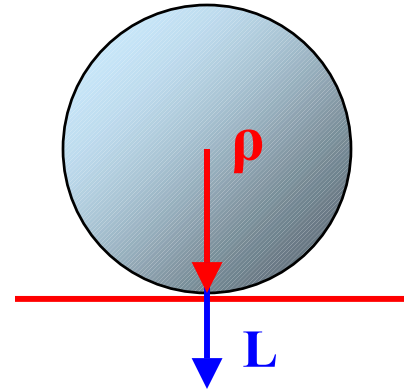
Invariable Plane

- Inertia ellipsoid rolls on the invariable plane
 - Rotation of ellipsoid gives the rotation of the body
 - Direction of $\mathbf{p}$ gives the direction of $\boldsymbol{\omega}$ in space
- Touching point draws curves
 - Curve drawn on the ellipsoid = **polhode**
 - Curve drawn on the invariable plane = **herpolhode**
- Let's examine a few simple cases



Simple Cases

- Inertia ellipsoid is a sphere ($I_1 = I_2 = I_3$)
 - $\boldsymbol{\rho}$ is constant and parallel to $\mathbf{L}$
 - Stable rotation



Simple Cases

- Initial axis is close to one of the principal axes

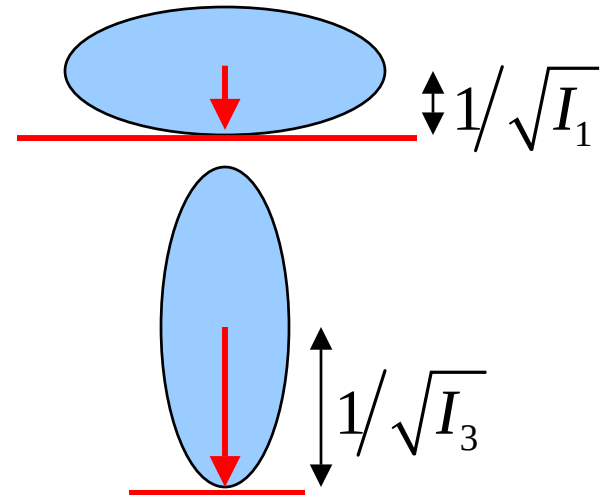
- Assume $I_1 > I_2 > I_3$
- Stable rotation around I_1 and I_3
- Not so obvious around I_2
 - If $\omega_1 = \omega_3 = 0$, ω_2 is constant

$$I_1 \dot{\omega}_1 - \omega_2 \omega_3 (I_2 - I_3) = 0$$

$$I_2 \dot{\omega}_2 - \omega_3 \omega_1 (I_3 - I_1) = 0$$

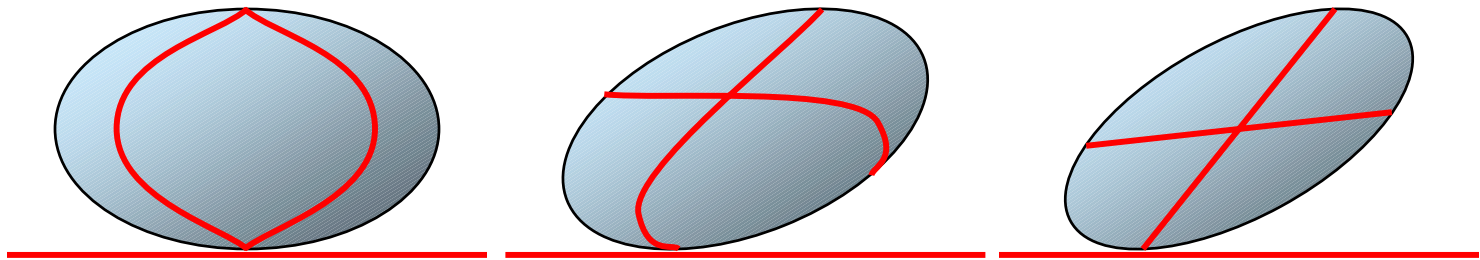
$$I_3 \dot{\omega}_3 - \omega_1 \omega_2 (I_1 - I_2) = 0$$

- Small deviation leads to instability



Simple Cases

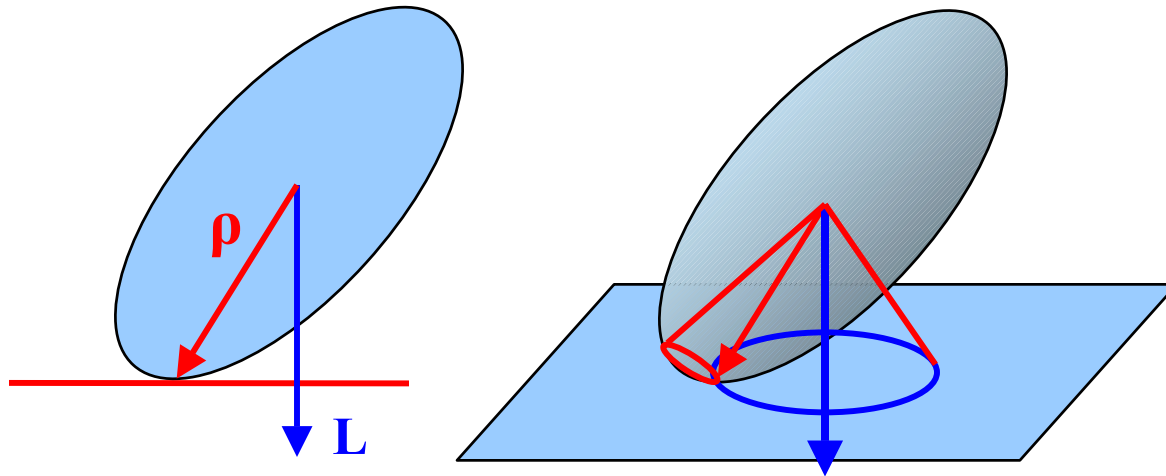
- Since $I_1 > I_2 > I_3$, distance $1/\sqrt{I_2}$ allows a polhode that wraps around the inertia ellipsoid



- Rotation around a principal axis is stable except for the one with the intermediate moment of inertia

Simple Cases

- Inertia ellipsoid is symmetric around one axis
 - $I_1 = I_2 \neq I_3$
 - ρ draws a cone (space cone) on the invariable plane
 - ρ draws a cone (body cone) in the inertia ellipsoid
 - Body cone rolls on the space cone



Precession

- $I_1 = I_2$ turns the Euler's equation of motion to

$$I_1 \dot{\omega}_1 = \omega_2 \omega_3 (I_1 - I_3)$$

$$I_1 \dot{\omega}_2 = \omega_3 \omega_1 (I_3 - I_1)$$

$$I_3 \dot{\omega}_3 = 0$$

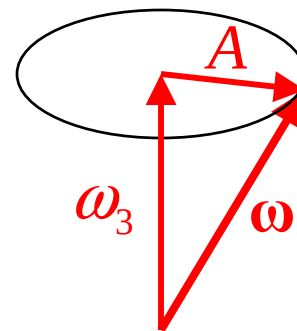
ω_3 is constant
Consider it as a given initial condition

$$\dot{\omega}_1 = -\Omega \omega_2, \quad \dot{\omega}_2 = \Omega \omega_1$$

$$\Omega \equiv \frac{I_3 - I_1}{I_1} \omega_3$$

$$\omega_1 = A \cos \Omega t, \quad \omega_2 = A \sin \Omega t$$

- ω **precesses** around the I_3 axis
 - Draws the body cone



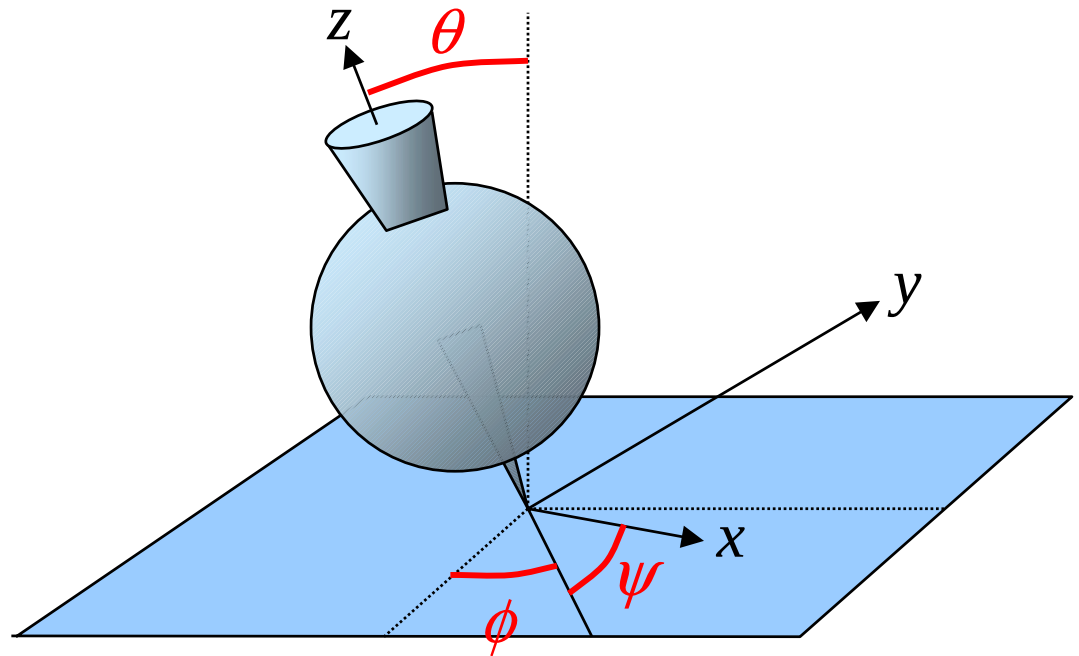
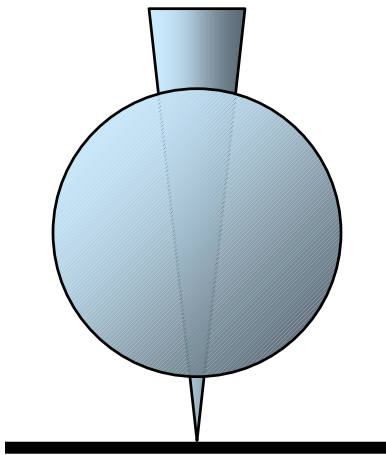
Rotation Under Torque

- We introduce torque
 - Things get messy
- Consider a spinning top
 - Define Euler angles

$$I_1 \dot{\omega}_1 - \omega_2 \omega_3 (I_2 - I_3) = N_1$$

$$I_2 \dot{\omega}_2 - \omega_3 \omega_1 (I_3 - I_1) = N_2$$

$$I_3 \dot{\omega}_3 - \omega_1 \omega_2 (I_1 - I_2) = N_3$$



Lagrangian

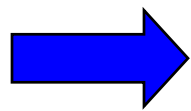
- Assume $I_1 = I_2 \neq I_3$

- Kinetic energy given by

$$T = \frac{1}{2} I_1 (\omega_1^2 + \omega_2^2) + \frac{1}{2} I_3 \omega_3^2$$

- Use Euler angles

$$\boldsymbol{\omega} = \begin{bmatrix} \dot{\phi} \sin \psi \sin \theta + \dot{\theta} \cos \psi \\ \dot{\phi} \cos \psi \sin \theta - \dot{\theta} \sin \psi \\ \dot{\phi} \cos \theta + \dot{\psi} \end{bmatrix}$$



$$T = \frac{I_1}{2} (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{I_3}{2} (\dot{\phi} \cos \theta + \dot{\psi})^2$$

Lagrangian

- Potential energy is given by the height of the CoM

$$V = Mgl \cos \theta$$

- Lagrangian is

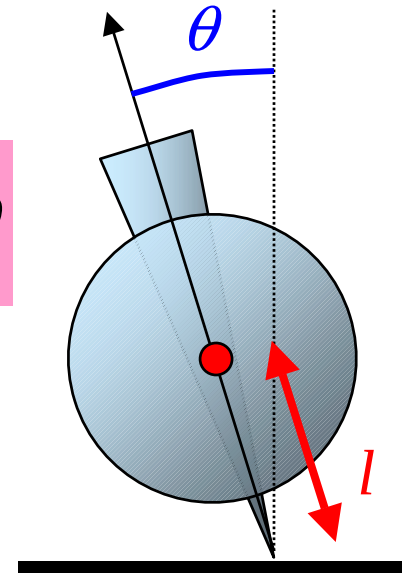
$$L = \frac{I_1}{2} (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{I_3}{2} (\dot{\phi} \cos \theta + \dot{\psi})^2 - Mgl \cos \theta$$

- Finally we are in real business!
- How we solve this?

- Note ϕ and ψ are cyclic

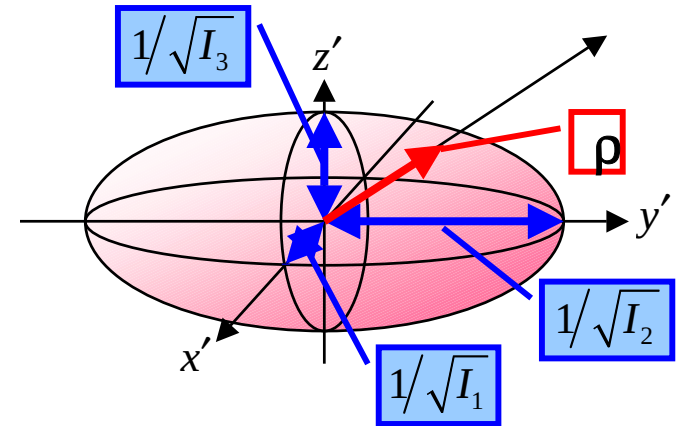
- Can define conjugate momenta that conserve

- To be continued ...



Summary

- Discussed rotational motion of rigid bodies
 - Euler's equation of motion
- Analyzed torque-free rotation
 - Introduced the inertia ellipsoid
 - It rolls on the invariant plane
 - Dealt with simple cases
- Started discussing heavy top
 - Found Lagrangian → Will solve this next time



$$L = \frac{I_1}{2}(\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{I_3}{2}(\dot{\phi} \cos \theta + \dot{\psi})^2 - Mgl \cos \theta$$